FlipFlop name	Characteristic Table			Characteristic Equation	Excitation Table			
SR	S	R	Qnext		Q	Qnext	S	R
	0	0		Qnext= S+R’Q	0	0		
	0	1			0	1		
	1	0			1	0		
	1	1		Where SR=0	1	1		
JK	J	K	Qnext		Q	Qnext	J	K
	0	0		Qnext= JQ’+K’Q	0	0		
	0	1			0	1		
	1	0			1	0		
	1	1			1	1		
D	D		Qnext		Q	Qnext	D	
	0			Qnext=D	0	0		
	1				0	1		
					1	0		
					1	1		
T	T		Qnext		Q	Qnext	T	
	0			Qnext= TQ’+T’Q	0	0		
	1				0	1		
					1	0		
					1	1		

S-R Flip Flop to D Flip Flop

Conversion Table

D Input	Outputs Qp	Outputs Qp+1	S-R Inputs S	S-R Inputs R
0	0	0	0	X
0	1	0	0	1
1	0	1	1	0
1	1	1	X	0

K-maps

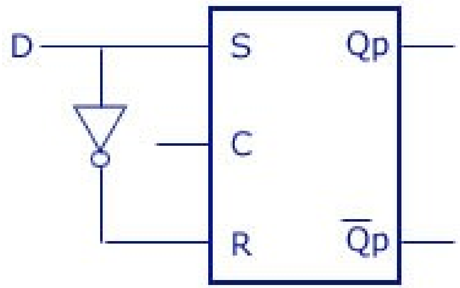
D \ Qp	0	1
0	0	0
1	$\boxed{1}$	$\boxed{X}$

$S = D$

D \ Qp	0	1
0	$\boxed{X}$	$\boxed{1}$
1	0	0

$R = \overline{D}$

Logic Diagram



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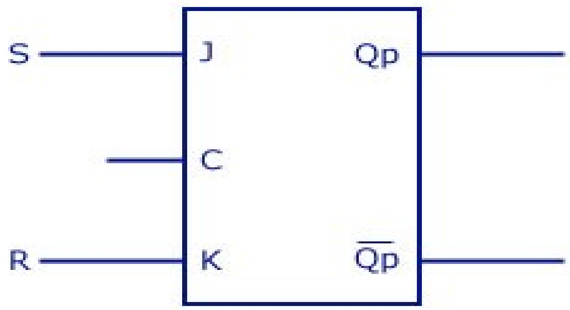
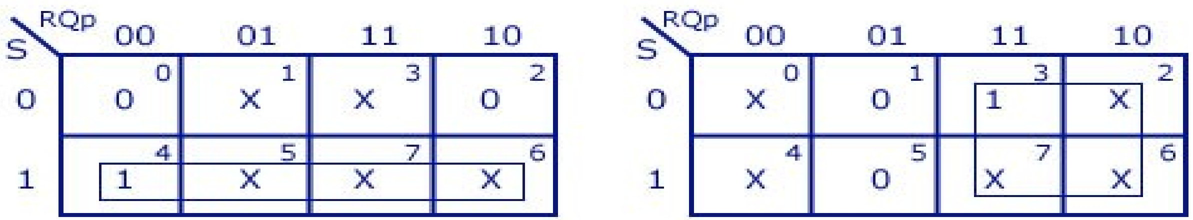
J-K Flip Flop to S-R Flip Flop

Conversion Table

S-R Inputs S	R	Outputs Qp	Qp+1	J-K Inputs J	K
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	X	1
1	0	0	1	1	X
1	0	1	1	X	0
1	1	Invalid		Dont care	
1	1	Invalid		Dont care	

Logic Diagram





J=S

K-maps

K=R

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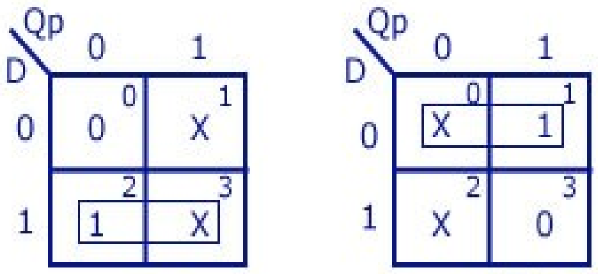
J-K Flip Flop to D Flip Flop

Conversion Table

D Input	Outputs		J-K Inputs	
	Qp	Qp+1	J	K
0	0	0	0	X
0	1	0	X	1
1	0	1	1	X
1	1	0	X	0

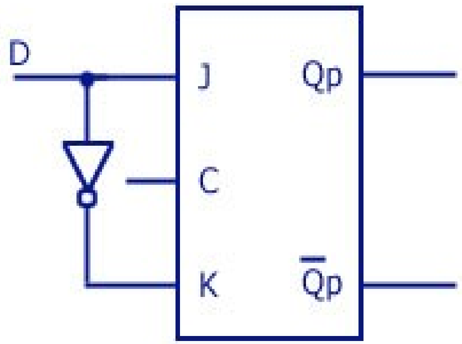
K-maps



J=D

$K=\overline{D}$

Logic Diagram



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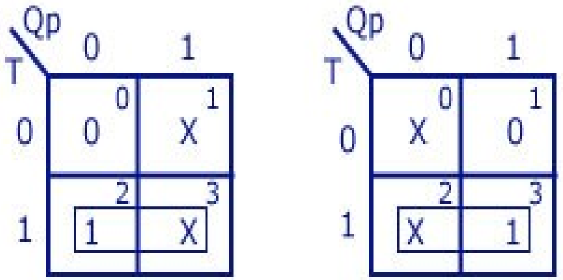
J-K Flip Flop to T Flip Flop

Conversion Table

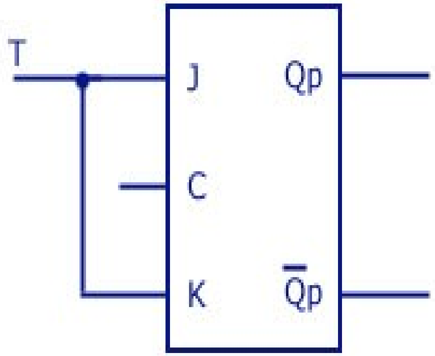
T Input	Outputs		J-K Inputs	
	Qp	Qp+1	J	K
0	0	0	0	X
0	1	1	X	0
1	0	1	1	X
1	1	0	X	1

K-maps



J=T K=T

Logic Diagram

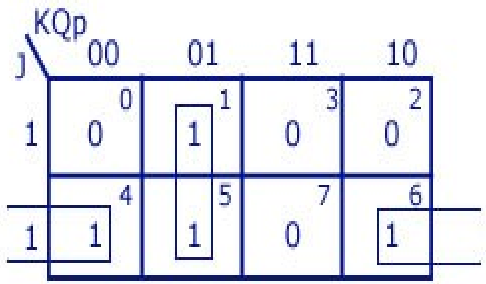


D Flip Flop to J-K Flip Flop

Conversion Table

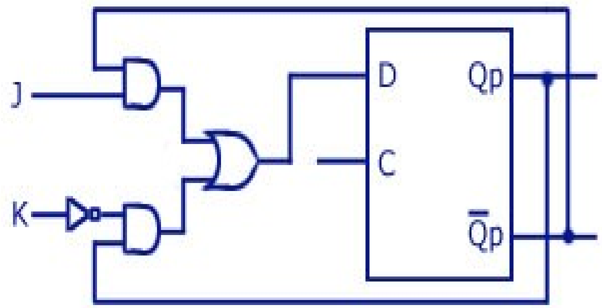
J-K Input: J	K	Outputs: Qp	Qp+1	D Input
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

K-map



$$D = J\overline{Qp} + \overline{K}Qp$$

Logic Diagram



D Flip Flop to S-R Flip Flop

Conversion Table

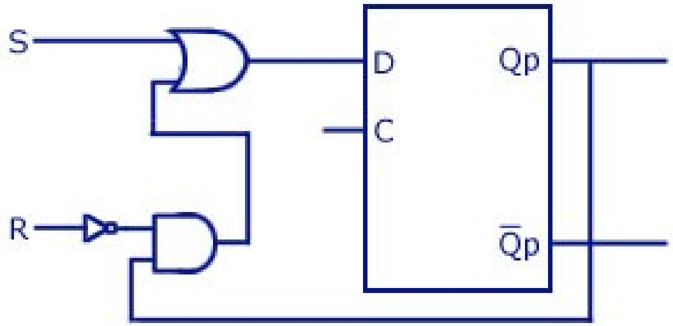
S-R Inputs		Outputs		D Input
S	R	Qp	Qp+1	
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	Invalid		Dont care
1	1	Invalid		Dont care

K-map

S \ RQp	00	01	11	10
0	0 (0)	1 (1)	0 (3)	0 (2)
1	1 (4)	1 (5)	X (7)	X (6)

$$D = S + \bar{R}Q_n$$

Logic Diagram



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